

# Real Analysis I

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## Contents

### Homework 2: Week 2 Due Sep 9

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I use the following lemma to prove Exercise 3.J.

**Lemma 1** (pigeonhole principle). Let  $Y$  be a finite set and  $X$  be finite with cardinality greater than  $|Y|$ . Then for any function  $f : X \rightarrow Y$ , there exists  $x_1, x_2 \in X$  such that  $f(x_1) = f(x_2)$ .

*Proof.* Denote  $A_y$  by the set  $\{x \in X : f(x) = y\}$ . We must have  $|A_y| < 2$  for all  $y \in Y$ , otherwise there will be more than one element from  $X$  mapped to the same element in  $Y$ . Hence  $0 \leq |A_y| \leq 1$ . We have

$$\left| \bigcup_{y \in Y} A_y \right| = \sum_{y \in Y} |A_y| \leq \sum_{y \in Y} 1 = |Y|.$$

The first equality comes from the fact that  $A_y$  are all disjoint and finite (since elements mapped to different targets must be different).

But for any  $x \in X$ , it is mapped to some element  $y = f(x)$ . In other words,  $x \in A_{f(x)} \subset \bigcup_{y \in Y} A_y$ . In the other direction, for any  $x \in \bigcup_{y \in Y} A_y$ , it is some element in  $X$  by definition. We have shown that  $X = \bigcup_{y \in Y} A_y$  and therefore  $|X| = \left| \bigcup_{y \in Y} A_y \right| \leq |Y|$  which is a contradiction to the assumption that  $|X| > |Y|$ . Therefore there must exist some  $y$  such that  $|A_y| \geq 2$ , which is exactly what we want to prove.  $\square$

I prove this exercise in extra, because I used it in Exercise 7.I.

7.I. Show that the collection of "right hand" end points in  $\mathbf{F}$  is denumerable. Show that if all these end points are deleted from  $\mathbf{F}$ , then what remains can be put onto one-one correspondence with all of  $[0, 1)$ . Conclude that the set  $\mathbf{F}$  is not countable.

*Proof.* Each "right hand" end point is in the form  $\frac{k+1}{3^n}$  where  $k, n \in \mathbb{N}$ . Therefore it is a rational number, and the set of rational numbers  $\mathbb{Q}$  is denumerable.

Because the set of right hand end points is a subset of rational numbers, it is denumerable.

Now we construct cascaded segmentation of the half-open interval  $[0, 1)$ . For each nonnegative integer  $n$  and  $0 \leq k \leq n$ , define  $A_{nk}$  to be

$$A_{nk} = [\frac{k}{2^n}, \frac{k+1}{2^n}).$$

We use induction to verify the following properties:

1. for all  $n$  and all distinct  $k, l$ ,

$$A_{nk} \cap A_{nl} = \emptyset.$$

2. for all  $n$  there are exactly  $2^n$  disjoint sets of the form  $A_{nk}$ .

3. for all  $n$ ,

$$\bigcup_{0 \leq k \leq n} A_{nk} = [0, 1).$$

4. Also, for any  $n, k$ ,  $A_{n+1, l} \subset A_{nk}$  if and only if

$$2k \leq l \leq 2k + 1.$$

Now define the sequence  $(x_n)_{n=0}^\infty$  as

$$x_0 = \begin{cases} 0 & x \in A_{00} \\ 1 & x \in A_{01} \end{cases}$$

$$x_n = \begin{cases} 0 & x \in A_{n, 2k} \text{ where } x \in A_{n-1, k} \\ 1 & x \in A_{n, 2k+1} \text{ where } x \in A_{n-1, k} \end{cases}$$

Now the function  $F$  defined below is a bijection between all 0-1 sequences that are not fully 1 and  $[0, 1)$ :

$$F : \{0, 1\}^\omega \setminus (1)_{n \in \mathbb{N}} \longrightarrow [0, 1)$$

$$(x_n)_{n \geq 0} \longmapsto F((x_n)_{n \geq 0}) = \sum_{n=0}^{\infty} \frac{x_n}{2^{n+1}}.$$

Now we introduce a function  $G$ :

$$G : \{0, 1\}^\omega \setminus (1)_{n \in \mathbb{N}} \longrightarrow [0, 1)$$

$$(x_n)_{n \geq 0} \longmapsto G((x_n)_{n \geq 0}) = \sum_{n=0}^{\infty} \frac{2x_n}{3^{n+1}}.$$

We verify that  $G$  is a bijection between all not-all-one 0-1 sequences and the cantor set. Then we have a bijection between  $[0, 1)$  and the cantor set.  $\square$

6.M, 7.F, 7.G, 7.J, 7.L, 3.F, 3.J

1. 6.M. Modify the argument given in Theorem 6.8 to show that if  $a > 0$ , then the number

$$b = \sup \{y \in \mathbf{R} : 0 \leq y, \quad y^2 \leq a\}$$

exists and has the property that  $b^2 = a$ . This number will be denoted by  $\sqrt{a}$  or  $a^{1/2}$ , and is called the positive square root of  $a$ .

*Proof.* Denote the set by  $S$ . We first show that the supremum exists. Clearly  $0 \in S$ , so the set is nonempty. Next we show that the set is bounded from above. We split into cases where  $x \leq 1$  and  $x > 1$ . When  $x \leq 1$ , we claim that the set is bounded above by 1. Suppose not, then there exists some  $s \in S$  such that  $s > 1$ , hence  $s^2 > 1$  but we have  $s^2 \leq a \leq 1$ , a contradiction. When  $x > 1$ , we claim that the set is bounded above by  $a$ . Suppose not, then there exists some  $s \in S$  such that  $s > a$ , hence  $s^2 > a^2$  but we have  $s^2 \leq a < a^2$  (the second inequality is due to  $a > 1$ ), a contradiction. In both cases the set is bounded above. By supremum property of  $\mathbb{R}$ ,  $b = \sup S$  exists.

Now we show  $b^2 = a$ . Suppose not, then by trichotomy property of real ordering, we have either  $b^2 < a$  or  $b^2 > a$ . When  $b^2 < a$ , we take  $n \in \mathbb{N}$ , such that  $\frac{1}{n} < \frac{a-b^2}{2b+1}$  (using Corollary 6.7 (b) of Archimedean Property). In this case

$$(b + \frac{1}{n})^2 = b^2 + \frac{1}{n} \left(2b + \frac{1}{n}\right) \leq b^2 + \frac{1}{n} (2b + 1) < b^2 + (a - b^2) = a.$$

This means  $b + \frac{1}{n} \in S$ , contrary to the fact that  $b$  is an upper bound of  $S$ .

When  $b^2 > a$ , we take  $n \in \mathbb{N}$  such that  $\frac{1}{n} < \frac{b^2-a}{2b}$  (such  $n$  exists by Corollary 6.7(b) of the Archimedean Property). Since  $b = \sup S$ , there exists an  $s_0 \in S$  such that  $b - \frac{1}{n} < s_0$ . But this implies that

$$a < b^2 - 2\frac{b}{n} < b^2 - 2\frac{b}{n} + \frac{1}{n^2} = \left(b - \frac{1}{n}\right)^2 < s_0^2.$$

But this means  $s_0^2 > a$ , contrary to the fact that  $s_0 \in S$ .

We have excluded the possibilities of  $b^2 < a$  and  $b^2 > a$ , so the only possibility left is  $b^2 = a$ .  $\square$

2. 7.F. Let  $J_n = (0, 1/n)$  for  $n \in \mathbb{N}$ . Show that this sequence of intervals is nested, but that there is no common point.

*Proof.* We first show the sequence is nested. Fix some  $n \in \mathbb{N}$ . It suffices to show that  $J_{n+1} \subset J_n$ . Take some  $x \in J_{n+1}$ , then by definition we have  $0 < x < \frac{1}{n+1}$ . But since  $\frac{1}{n+1} < \frac{1}{n}$ , we have  $0 < x < \frac{1}{n}$ , hence by definition again we have  $x \in J_n$ .

Now we show there is no common point. Fix any  $x \in J_1 = (0, 1)$ . By Corollary 6.7(b), there exists some  $n \in \mathbb{N}$  such that  $0 < \frac{1}{n} < x$ . In other words,  $x \notin J_n$ . Therefore,  $\bigcap_{n \in \mathbb{N}} J_n = \emptyset$  since each element is not in at least one of the sets.  $\square$

3. 7.G. If  $I_n = [a_n, b_n]$ ,  $n \in \mathbf{N}$ , is a nested sequence of closed cells, show that

$$a_1 \leq a_2 \leq \cdots \leq a_n \leq \cdots \leq b_m \leq \cdots \leq b_2 \leq b_1.$$

If we put  $\xi = \sup \{a_n : n \in \mathbf{N}\}$  and  $\eta = \inf \{b_m : m \in \mathbf{N}\}$ , show that  $[\xi, \eta] = \bigcap_{n \in \mathbf{N}} I_n$ .

*Proof.* Take  $n, m \in \mathbf{N}$  where  $n \leq m$ . By assumption  $I_m \subset I_n$ . We need to show that  $a_n \leq a_m$  and  $b_m \leq b_n$ . By definition of closed intervals, for all  $x \in \mathbb{R}$ ,  $x \in I_n \Leftrightarrow a_n \leq x \leq b_n$  and  $x \in I_m \Leftrightarrow a_m \leq x \leq b_m$ . By set inclusion,  $x \in I_m \Rightarrow x \in I_n$ . Therefore

$$a_m \leq x \leq b_m \Rightarrow a_n \leq x \leq b_n.$$

Suppose that  $a_m < a_n$ . Then there exists some  $x$  such that  $a_m < x < a_n$ , such that  $a_n \leq x \leq b_n$  is not true, therefore the implication is not true for the specific  $x$ , a contradiction to the universally quantified statement. By trichotomy of real order, we have  $a_m \geq a_n$ . By the same reasoning we have  $b_m \leq b_n$ .

Now we show that  $[\xi, \eta] = \bigcap_{n \in \mathbf{N}} I_n$ . We first show that  $[\xi, \eta] \subset \bigcap_{n \in \mathbf{N}} I_n$ . Take  $x \in [\xi, \eta]$ , and take an arbitrary  $n \in \mathbf{N}$ . It suffices to show that  $x \in I_n$ . Since  $\xi$  is a supremum, we must have  $\xi \geq a_n$ . Therefore  $x \geq \xi \geq a_n$ . Similarly we have  $x \leq b_n$ . By definition of  $I_n$ , we have  $x \in I_n$ .

Next we show that  $[\xi, \eta] \supset \bigcap_{n \in \mathbf{N}} I_n$ . Take  $x \in \bigcap_{n \in \mathbf{N}} I_n$ . Then  $x \in I_n$  for all  $n \in \mathbf{N}$ . Suppose that  $x < \xi$ . Then there exists some  $m \in \mathbf{N}$  such that  $x < a_m \leq \xi$ . Clearly  $x \in I_m$  is not true, a contradiction. Therefore we must have  $x \geq \xi$ . By the same reasoning we have  $x \leq \eta$ , so  $x \in [\xi, \eta]$ .  $\square$

4. 7.J. Every open interval  $(a, b)$  which contains a point of  $\mathbf{F}$  also contains an entire "middle third" set which belongs to  $\mathcal{C}(\mathbf{F})$ . Hence  $\mathbf{F}$  does not contain any non-void open interval.

*Proof.* Fix some interval  $(a, b) \subset \mathbb{R}$  and some point  $x \in \mathbf{F} \cap (a, b)$ . Since  $a < x < b$ , we have  $b - x > 0$  and  $x - a > 0$ . We can choose some  $n \in \mathbf{N}$  such that

$$3^n > \max\left(\frac{1}{x-a}, \frac{1}{b-x}\right).$$

To justify this, note that by Bernoulli's Inequality

$$3^n = (1+2)^n \geq 1+2n$$

and by Archimedean property we can choose  $n$  such that  $n > \frac{\max(\frac{1}{x-a}, \frac{1}{b-x})-1}{2}$ .

By Corollary 6.7(c) of the Archimedean property, there exists  $k \in \mathbf{N}$  such that  $k \leq 3^n x < k+1$ . Since  $3^n x - 3^n a > 1$ , we have  $k > 3^n x - 1 > 3^n a$ . Since  $3^n b - 3^n x > 1$ , we have  $k+1 \leq 3^n x + 1 < 3^n b$ . Therefore  $[\frac{k}{3^n}, \frac{k+1}{3^n}] \subset (a, b)$ .

By definition and construction,  $x \in [\frac{k}{3^n}, \frac{k+1}{3^n}] \subset F_n$ .  $F_{n+1}$  takes the middle third from this interval, which is  $[\frac{3k}{3^{n+1}}, \frac{3k+1}{3^{n+1}}] \cup [\frac{3k+2}{3^{n+1}}, \frac{3k+3}{3^{n+1}}] \subset F_{n+1}$ . The middle-third  $M = ([\frac{3k+1}{3^{n+1}}, \frac{3k+2}{3^{n+1}}])$  satisfies the properties  $M \subset F_n$  and  $M \subset (a, b)$ .  $\square$

5. 7.L. Show that  $\mathbf{F}$  is not the union of a countable collection of closed intervals.

*Proof.* Suppose  $\mathbf{F} = \bigcup_{n \in \mathbb{N}} I_n$  where each  $I_n = [a_n, b_n]$ . Suppose there exists some  $m \in \mathbb{N}$  such that  $a_m < b_m$ . Then the open interval  $(a_m, b_m)$  is nonempty, and has a nonempty intersection with  $\mathbf{F}$ . By Exercise 7.J, the interval  $(a_m, b_m)$  contains a "middle-third" set. This contradicts the fact that  $[a_n, b_n] \subset \mathbf{F}$ .

Therefore  $a_n \geq b_n$  for all  $n \in \mathbb{N}$ . If  $a_n > b_n$  then the interval is empty, so we have  $I_n = \{x_n\} \subset \mathbf{F}$  for at most countable number of  $I_n$ , and the rest of them are empty. This is to say that the set  $\mathbf{F}$  is at most countable. But we have concluded in Exercise 7.I that  $\mathbf{F}$  is not countable.  $\square$

6. 3.F. Use the fact that every infinite set has a denumerable subset to show that every infinite set can be put into one-one correspondence with a proper subset of itself.

*Proof.* Consider an infinite set  $X$  and a denumerable subset  $A \subset X$ . Denumerate  $A$  by  $a_1, a_2, \dots, a_i, \dots$ , and define the map

$$F : X \longrightarrow X \setminus \{a_1\}$$

$$x \longmapsto F(x) = \begin{cases} x & x \notin A \\ a_{i+1} & x = a_i \in A \end{cases}.$$

$F$  is a function because for every  $x$  either  $x \notin A$ , or  $x \in A$ , in this case there must exist  $i \in \mathbb{N}$  such that  $x = a_i$ .  $F$  is injective since for any  $x_1 \neq x_2$ , either they both  $\notin A$ , in this case  $F(x_1) = x_1 \neq x_2 = F(x_2)$ ; or they are both  $\in A$ , in this case  $F(x_1) = F(a_i) = a_{i+1} \neq a_{j+1} = F(a_j) = F(x_2)$ ; or one of them is in  $A$  and the other is not, in this case the claim is obvious.

$F$  is surjective because for any  $x \in X \setminus A$ , its preimage is itself; and for any  $x = a_i \in A \setminus \{a_1\}$  where  $i \neq 1$ , its preimage is  $a_{i-1} \in A \setminus \{a_1\}$ .  $\square$

7. 3.J. Prove that  $\mathbb{N}$  cannot be put into one-one correspondence with any initial segment of  $\mathbb{N}$ .

*Proof.* Suppose for the sake of contradiction that there exists some  $n \in \mathbb{N}$  such that there is a bijection  $f : \mathbb{N} \rightarrow S_n$ . Hence in particular  $f$  is injective. Then by restricting the domain,  $f|_{S_{n+1}} : S_{n+1} \rightarrow S_n$  is injective. However this cannot be true since by the pigeonhole principle there must exist two elements in  $S_{n+1}$  that are mapped to the same element in  $S_n$ .  $\square$